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Distributed Email Service

Modern business communication heavily relies on email, with billions of users worldwide using it for work and personal matters. While seemingly simple, the underlying system for a large-scale email service is complex, encompassing various components and protocols to ensure reliable, secure, and efficient delivery. This chapter delves into the design of such a distributed email service.

Gmail Outlook Yahoo Mail

Step 1 - Understand the Problem and Establish Design Scope

Email protocols are fundamental communication standards that govern how emails are sent, received, and managed across different systems. Key protocols include SMTP (Simple Mail Transfer Protocol) for sending emails, POP3 (Post Office Protocol version 3) and IMAP (Internet Message Access Protocol) for retrieving emails, and various extensions and security protocols like TLS (Transport Layer Security) for encryption.

Candidate: What are the most important features for a large-scale email service?

Interviewer: Focus on the following core functionalities:

- Sending emails
- Receiving emails
- Storing mail data
- Retrieving mail data
- Filtering spam and malicious content
- Ensuring data encryption and security

Candidate: What scale are we dealing with? Should we consider high availability and low latency?

Interviewer: Assume the system should handle hundreds of millions of users. Yes, high availability is important, but we can discuss latency requirements later.

Candidate: What about email formats? Do we need to support multimedia attachments?

Interviewer: Initially, focus on text-based emails. We can discuss multimedia attachments later if time allows.

Candidate: Should we consider third-party integrations, like calendar or task management?

Interviewer: No, that's beyond the scope for this design.

Candidate: Do we need to support real-time email delivery?

Interviewer: Near real-time is acceptable, but we want to minimize delivery time.

Candidate: What about data retention? How long should we store emails?

Interviewer: Assume we need to store emails for 10 years.

Candidate: Do we need to consider disaster recovery and data backup?

Interviewer: Yes, that's important for ensuring data reliability.

Candidate: What about spam and malicious content? Should we implement filtering?

Interviewer: Yes, spam filtering and malware detection are important features.

Candidate: Do we need to support different email formats and protocols?

Interviewer: Focus on the standard SMTP, POP3, and IMAP protocols for now.

Candidate: Should we consider mobile access?

Interviewer: Yes, mobile access is important, but we can discuss the details later.

Candidate: Do we need to implement email encryption?

Interviewer: Yes, encryption is important for security.

Candidate: Should we consider email threading or conversation grouping?

Interviewer: Not for this initial design.

Candidate: Do we need to support email search functionality?

Interviewer: Yes, efficient email search is important.

Candidate: Should we consider spam reporting and user feedback mechanisms?

Interviewer: Yes, that would be a good feature.

Candidate: Do we need to consider email archiving?

Interviewer: Yes, archiving is important for long-term storage.

Candidate: Should we consider email quotas for users?

Interviewer: Yes, implementing quotas is a good idea.

Candidate: Do we need to support email aliases?

Interviewer: Not for this initial design.

Candidate: Should we consider email templates?

Interviewer: Not for this initial design.

Candidate: Do we need to support email scheduling?

Interviewer: Not for this initial design.

Candidate: Should we consider email analytics?

Interviewer: Not for this initial design.

Candidate: Do we need to consider email compliance (e.g., GDPR, HIPAA)?

Interviewer: Yes, compliance with relevant regulations is important.

Candidate: Should we consider email deliverability optimization?

Interviewer: Yes, ensuring high deliverability is important.

Candidate: Do we need to consider email authentication (SPF, DKIM, DMARC)?

Interviewer: Yes, email authentication is important for security.

Candidate: Should we consider email threat intelligence?

Interviewer: Not for this initial design.

Candidate: Do we need to consider email encryption at rest?

Interviewer: Yes, encryption at rest is important for data security.

Candidate: Should we consider email compression?

Interviewer: Not for this initial design.

Candidate: Do we need to consider email deduplication?

Interviewer: Not for this initial design.

Candidate: Should we consider email prioritization?

Interviewer: Not for this initial design.

Candidate: Do we need to consider email threading?

Interviewer: Not for this initial design.

Candidate: Should we consider email templates?

Interviewer: Not for this initial design.

Candidate: Do we need to consider